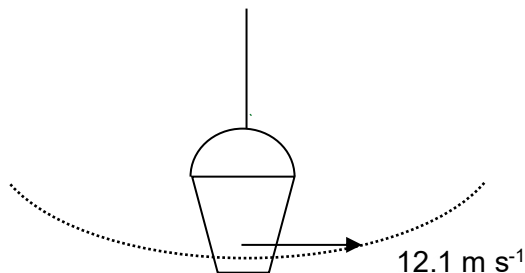


- 10 A 1.2 kg bucket is held by a string and whirled in a vertical circle. The radius of the circle is 1.5 m. The speed of the bucket is 12.1 m s^{-1} at the bottom of the loop and subsequently, the bucket reached the top of the circle.



At the bottom of the
loop

What is the difference in the tension in the string at these two positions?

A 58 N

B 71 N

C 129 N

D 187 N

By COE,

$$\frac{1}{2} m(12.1)^2 - \frac{1}{2} m v^2 = mg(3.0)$$

$$V \text{ at the top} = 9.36 \text{ ms}^{-1}$$

$$\text{Tension at the top : } T_1 = mv^2 / r - mg = (1.2)(9.36)^2 / 1.5 - 1.2 (9.81) = 58.3 \text{ N}$$

$$\text{Tension at the bottom, } T_2 = mv^2 / r + mg = (1.2)(12.1)^2 / 1.5 + 1.2 (9.81) = 129 \text{ N}$$

$$\text{Difference} = 71 \text{ N}$$

Answer : B